Number Detectives — multiples, factors and patterns ⏱ ~25 min

NAME _____ DUE _____ SCORE / EFFORT _____

1 · This week

1. a) Write the multiples of 8 up to 80:

b) Which common multiples of 8 and 12 are less than 100?

2. List all the factor pairs of 40:

3. The common factors of 18 and 27 are _____. The greatest common factor is _____.

4. Divisibility. Tick (✔) each box that works. Use the tricks — no dividing!

Number	Digit sum	÷ 2	÷ 3	÷ 5	÷ 9	÷ 10
270						
1 035						
4 128						

5. Patterns. Fill in the missing terms and write the rule.

a) 1⅔, 2, 2⅓, _____, _____ Rule: _____

b) 4.75, 4.5, 4.25, _____, _____ Rule: _____

c) 0.35, 0.4, _____, _____, 0.55 Rule: _____

6. ⭐⭐⭐ **Paving a courtyard.** A landscaper has **36** square paving stones (each 1 m by 1 m). She wants to lay all of them in one solid rectangle.

a) List every rectangle she could make and its perimeter.

Length (m)	Width (m)	Perimeter (m)

b) She will put a garden edging around the outside. Which rectangle needs the **least** edging? _____

7. ⭐⭐⭐ **Prime time.** a) Is 87 prime or composite? Prove it.

b) Find the two prime numbers between 80 and 90. _____ _____

2 · Spaced revision (Terms 1–3)

8. *(Week 11)* 347 × 6 = _____ **9.** *(Week 12)* 845 ÷ 4 = _____

WORKING

10. *(Week 9)* A movie starts at 6:40 pm and runs for 1 hour 55 minutes. What time does it finish?

_____

11. *(Week 14)* Order from smallest to largest: 0.7 · 0.07 · 0.71 · 0.17

12. *(Week 23)* You face north and make a three-quarter turn clockwise. Which way are you facing now?

_____

13. *(Week 26)* When you roll a fair dice (1 to 6), is a prime number or a composite number more likely? Explain.

3 · At home

14. Calendar multiples. Use a calendar for December. *(Hypothetical plan.)* You water the vegie patch every **3** days and wash the dog every **4** days, and you do **both** on 1 December.
a) On which December dates do you do both jobs? _____
b) How many days apart are those dates? Why? _____

15. Explain to someone at home the difference between a **factor** and a **multiple**, using the number 12. Write what you said: